

Ladder MPS+Mean-Field Project: Living Methods Notes

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Abstract

These notes summarize the methods currently implemented in the modular `ladder_mps_mft` workflow. The calculation combines finite-system matrix-product-state (MPS) ground states for an interacting two-leg ladder with a perturbative transverse mean-field (MF) map, explicit density targeting, fixed-point and physical-orbit validation, and a zero-temperature variational energy comparison. This is an implementation-oriented living document, not a claim of thermodynamic convergence. Unless stated otherwise, “accepted” means numerically self-consistent at the simulated length and bond dimension.

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1 Scope and notation

The authoritative implementation is the Julia package in `ladder_mps_mft/src`. The monolithic scripts at the repository root are retained for legacy provenance but are not the definition of the current method.

The simulated object is one open two-leg ladder with L rungs and $2L$ physical sites. A site is denoted $a = (i, \ell)$, with rung $i = 1, \dots, L$, leg $\ell \in \{0, 1\}$, and spin $\sigma \in \{\uparrow, \downarrow\}$. The MPS ordering is rung-major, $(1, 0), (1, 1), (2, 0), (2, 1), \dots$. The stored density is per physical site,

$$n = \frac{N}{2L}, \quad N_\star = 2Ln_\star. \quad (1)$$

All present MF fields and measured correlators are restricted to their real parts.

2 Isolated-ladder Hamiltonian

The bare ladder Hamiltonian is the extended Hubbard model

$$\begin{aligned}
H_{\text{lad}} = & -t \sum_{i=1}^{L-1} \sum_{\ell, \sigma} \left(c_{i\ell\sigma}^\dagger c_{i+1, \ell\sigma} + \text{H.c.} \right) - t_0 \sum_{i=1}^L \sum_{\sigma} \left(c_{i0\sigma}^\dagger c_{i1\sigma} + \text{H.c.} \right) \\
& + U \sum_{i, \ell} n_{i\ell\uparrow} n_{i\ell\downarrow} + V \sum_{\langle a, b \rangle_{\text{lad}}} n_a n_b.
\end{aligned} \quad (2)$$

Here t is the leg hopping, t_0 the rung hopping, and the V sum contains nearest-neighbor leg and rung bonds. Open boundary conditions are used along the ladder. The inter-ladder hopping $t_\perp$ is not inserted as an explicit third spatial direction; it enters the effective MF map at second order through $t_\perp^2/|E_p|$.

2.1 Pair-binding scale and registry

The hole pair-binding convention used by the registry is

$$E_p^{(h)} = E(N-2, 2S_z=0) + E(N, 2S_z=0) - 2E(N-1, 2S_z=1). \quad (3)$$

A negative value denotes a bound pair. The MF denominator is positive,

$$g = \frac{t_\perp^2}{|E_p|}, \quad p = 2g. \quad (4)$$

By default the code refuses an unbound or zero E_p . It first seeks an exact registry match in (L, U, V, t_0, n) , choosing the largest available bond dimension and then the smallest recorded relative discrepancy. Optional interpolation is linear in the *signed* E_p , only between bracketing t_0 values with identical (L, U, V, n) . Extrapolation and interpolation across a sign change are forbidden. The selected row, endpoints, weights, and registry hash are stored with every result.

The inexpensive check $t_\perp < |E_p|$ is necessary but not a complete weak-coupling test. Where available, $t_\perp$ is also compared with the isolated-ladder spin and charge gaps defined in Sec. 7.

3 Transverse mean-field construction

3.1 Fields, correlators, and effective Hamiltonian

The correlators F, X, n describe the state of a ladder, whereas the fields α, β, ν are the coefficients multiplying operators in its effective Hamiltonian. They are related by the transverse interaction, but are not interchangeable quantities: the correlators are dimensionless and the fields have units of energy. In particular, F measures anomalous pairing, X measures normal one-particle coherence, and n measures occupation; α, β, ν are their respective conjugate MF fields.

At SCF step k , collect the applied fields into $x_k = (\alpha, \beta, \nu)_k$:

- $\alpha_{ii'}^{\ell\ell'}$: anomalous singlet-pair field;
- $\beta_{\sigma,ii'}^{\ell\ell'}$: spin-resolved normal exchange field, with the on-site entry removed; and
- $\nu_{\sigma,il}$: spin-resolved Hartree field, stored in code as `mu_cdw` and distinct from the global chemical potential.

Only pairs with $|i - i'| \leq r_{\text{range}}$ are retained. Define the real correlators

$$F_{i'\ell',il} = \text{Re} \langle c_{i'\ell'\uparrow} c_{il\downarrow} \rangle, \quad (5)$$

$$X_{a,b}^\sigma = \text{Re} \langle c_{a\sigma}^\dagger c_{b\sigma} \rangle, \quad n_a^\sigma = \langle n_{a\sigma} \rangle. \quad (6)$$

Schematically, the fields produced by these measurements are

$$\alpha^{\text{out}} = K_{\text{pair}} F, \quad \beta^{\text{out}} = K_{\text{ex}} X, \quad \nu^{\text{out}} = K_{\text{dens}}(n - \tfrac{1}{2}). \quad (7)$$

Here each K denotes the appropriate linear map over spatial and, where applicable, spin indices, including the geometry, coupling scale $t_{\perp}^2/|E_p|$, and channel factors. The component formulas below specify the index orientations and excluded entries; the implementation also applies its configured field threshold. Thus α is not an additional independent pairing observable alongside F : it is the pairing source generated from the partner's correlators. The same distinction holds for β versus X and ν versus n .

The effective single-ladder problem is

$$H_{\text{eff}}[x_k, \mu_k] = H_{\text{lad}} - \mu_k N + H_{\text{lin}}[x_k], \quad (8)$$

with

$$\begin{aligned} H_{\text{lin}} = & - \sum_{ii'\ell\ell'} \alpha_{ii'}^{\ell\ell'} \left(c_{i'\ell'\uparrow} c_{i\ell\downarrow} + c_{i\ell\downarrow}^{\dagger} c_{i'\ell'\uparrow}^{\dagger} \right) \\ & + \sum_{ii'\ell\ell'\sigma} \beta_{\sigma,ii'}^{\ell\ell'} c_{i\ell\sigma}^{\dagger} c_{i'\ell'\sigma} + \sum_{i\ell\sigma} \nu_{\sigma,i\ell} n_{i\ell\sigma}. \end{aligned} \quad (9)$$

The computational sequence is therefore

$$x_k \longrightarrow \psi_k \longrightarrow (F_k, X_k, n_k) \longrightarrow y_k = (\alpha, \beta, \nu)_k^{\text{out}}. \quad (10)$$

The applied fields x_k are held fixed during the MPS solve (while the global μ_k is adjusted to target the density); the measured correlators then determine the outgoing fields y_k . A period-one fixed point requires $x_k = y_k$, not $\alpha = F$ or $\beta = X$. On a physical two-cycle, ladder A is solved in fields generated by ladder B and vice versa, so the current ladder's outgoing fields need not equal its applied fields. The spin- and site-dependent ν is fixed by the interaction map, whereas the single global μ enforces the total-density constraint. Their different roles remain even when ν has a uniform component.

3.2 Geometry-dependent map

Solving Eq. (8) and measuring F, X, n defines the raw map

$$y_k = f(x_k), \quad x_{k+1} = y_k \quad (11)$$

during an unmixed probe. The geometry labels are `cubic_frustrated`, `cubic_unfrustrated`, and `square`. For the same-leg anomalous fields, let $\mathbf{F} = (F_{i'0,i0}, F_{i'1,i1})^T$. Their exact implemented map is

$$\begin{pmatrix} \alpha_{ii'}^{00} \\ \alpha_{ii'}^{11} \end{pmatrix} = p M_G \mathbf{F}, \quad (12)$$

where

Geometry G	M_G	Cross-leg anomalous output
frustrated cubic	$\begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$	$(\alpha^{10}, \alpha^{01}) = 2p(F_{i'0,i1}, F_{i'1,i0})$
unfrustrated cubic	$\begin{pmatrix} 0 & 3 \\ 3 & 0 \end{pmatrix}$	$(\alpha^{10}, \alpha^{01}) = 2p(F_{i'1,i0}, F_{i'0,i1})$
square	$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$	$(\alpha^{10}, \alpha^{01}) = (0, 0)$

The normal fields β_σ use the same integer coefficient pattern with F replaced by the corresponding X^σ entries and the normal operator orientation in Eq. (9); their diagonal on-site entries are set to zero. This component-level map is implemented in `src/MeanField.jl`; changes to it require a simultaneous audit of the variational constants in Sec. 6.

Connection to the published ladder benchmark. The repulsive-ladder equations of Bollmark et al. [1], published Eqs. (51)–(58), match the `cubic_frustrated` map above: the same-leg kernel is $p\left(\begin{smallmatrix} 2 & 1 \\ 1 & 2 \end{smallmatrix}\right)$, and the cross-leg pairing and exchange terms have matching coefficients and index orientations. “Frustrated cubic” is our nomenclature. This identifies the connectivity underlying their Fig. 13 temperature benchmark. The dated source comparison and suggested manuscript wording are recorded in `docs/literature/SOURCE_NOTES.md`.

3.2.1 Bond-space interpretation of the pair and exchange fields

For each fixed ordered rung pair (i, i') , it is useful to regard

$$\alpha_{ii'} = (\alpha_{ii'}^{00}, \alpha_{ii'}^{11}, \alpha_{ii'}^{01}, \alpha_{ii'}^{10})^\top \quad (13)$$

as a four-component vector in leg-label space. The real exchange field β_σ has the analogous vector for each spin σ . The transverse kernel acts bond by bond: it mixes these four leg labels but never changes (i, i') . For either $q \in \{\alpha, \beta_\sigma\}$, the same-leg and cross-leg blocks decompose as

$$q_{ii'}^{\text{same}, \pm} = \frac{q_{ii'}^{00} \pm q_{ii'}^{11}}{\sqrt{2}}, \quad q_{ii'}^{\times, \pm} = \frac{q_{ii'}^{01} \pm q_{ii'}^{10}}{\sqrt{2}}. \quad (14)$$

The $+$ and $-$ combinations are respectively even and odd under reflection of the ladder, and hence carry transverse momenta $k_y = 0$ and $k_y = \pi$.

For the stored real singlet pair field and real Hermitian exchange field, the transpose relations are

$$\alpha_{ii'}^{\ell\ell'} = \alpha_{i'i}^{\ell'\ell}, \quad \beta_{\sigma, ii'}^{\ell\ell'} = \beta_{\sigma, i'i}^{\ell'\ell}. \quad (15)$$

Consequently, a cross-leg odd component obeys

$$q_{i'i}^{\times, -} = -q_{ii'}^{\times, -}. \quad (16)$$

It is therefore a legitimate diagonal-bond channel for $i \neq i'$, but it vanishes identically on a rung: $q_{ii}^{\times, -} = 0$. A cross-leg rung pair is necessarily leg-even, whereas the two oppositely oriented diagonals can form either an even or an odd combination. For β_σ , an imaginary antisymmetric rung amplitude would instead be a current/flux channel; complex fields of that kind are outside the present real-field implementation.

Leg-reflection parity is not the same as the usual s - versus d -wave form-factor label. In particular, the two leg bonds of a d -wave pair field have the same sign and are therefore leg-even; its d -wave character comes from the opposite sign of the rung-bond component, that is, from a relation between the same-leg and cross-leg blocks.

For each rung and spin, the Hartree update is

$$\begin{pmatrix} \nu_{\sigma, i0} \\ \nu_{\sigma, i1} \end{pmatrix} = K_G \left[\begin{pmatrix} n_{i0}^\sigma \\ n_{i1}^\sigma \end{pmatrix} - \frac{1}{2} \begin{pmatrix} 1 \\ 1 \end{pmatrix} \right], \quad (17)$$

with

$$K_G = \begin{cases} 2g \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}, & G = \text{frustrated cubic}, \\ 6g \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, & G = \text{unfrustrated cubic}, \\ 2g \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, & G = \text{square}. \end{cases} \quad (18)$$

The two-component Hartree field has the same even/odd leg-parity classification as the same-leg bond block. Table 1 consolidates the spectra of every real field in $x = (\alpha, \beta_\downarrow, \beta_\uparrow, \nu_\downarrow, \nu_\uparrow)$.

	Same-leg bond / local leg		Cross-leg bond	
	even, $k_y = 0$	odd, $k_y = \pi$	even, $k_y = 0$	odd, $k_y = \pi$
Applicable fields	$\alpha, \beta_\sigma, \nu_\sigma$	$\alpha, \beta_\sigma, \nu_\sigma$	α, β_σ	α, β_σ
frustrated cubic	$6g$	$2g$	$4g$	$4g$
unfrustrated cubic	$6g$	$-6g$	$4g$	$-4g$
square	$2g$	$-2g$	0	0

Table 1: Geometry-kernel eigenvalues k before multiplication by the isolated-ladder susceptibility and the channel-dependent source sign. The first two columns describe same-leg bonds for α, β_σ and the local leg vector for ν_σ . Cross-leg even modes can occupy rungs or diagonals; cross-leg odd modes require $i \neq i'$. The pair and exchange maps have the same geometry kernel, although their response signs differ. The exchange field omits the identical-site entry; the on-site singlet-pair entry is retained.

Both spin components of β_σ and ν_σ have the displayed spectrum. Their charge-like and spin-like combinations, $(q_\uparrow \pm q_\downarrow)/\sqrt{2}$, therefore inherit the same geometry eigenvalues, while their isolated-ladder susceptibilities need not be equal. The table exhausts the present real mean-field variables. The global chemical potential μ is a density-control Lagrange multiplier rather than a transverse mean-field channel; triplet pairing and imaginary current/flux fields are not implemented.

3.2.2 Kernel signs versus iteration signs

A kernel eigenvalue k is not by itself the eigenvalue of the complete linearized SCF map. If $\chi \geq 0$ denotes the magnitude of the isolated ladder's response in an unmixed eigenchannel, the source signs in Eq. (9) give

$$\lambda_\alpha = +k\chi_\alpha, \quad \lambda_\beta = -k\chi_\beta, \quad \lambda_\nu = -k\chi_\nu. \quad (19)$$

The anomalous term is written as minus the pair field times its conjugate operator, whereas the exchange and Hartree sources enter with plus signs. Kernel signs must therefore be read with opposite response conventions in the pair and normal one-body channels.

Equation (19) is scalar bookkeeping for an isolated mode. In general, the ladder response is a matrix: in particular, exchange and Hartree perturbations with the same symmetries can mix. The physical stability numbers are therefore the eigenvalues of the full Jacobian $J = \partial f / \partial x$, not the entries in Table 1. The table supplies the exact geometry part of that Jacobian.

What is and is not diagonal. Let $\mathbf{r}(x)$ denote all isolated-ladder correlators used to construct the next field. About a reference x_0 (which must be a fixed point for an exact stability analysis),

$$\delta \mathbf{r} = \mathcal{R} \delta x, \quad \mathcal{R}_{ab} = \left. \frac{\partial r_a}{\partial x_b} \right|_{x_0}, \quad J_G = K_G \mathcal{R}. \quad (20)$$

Here $\mathcal{R}$ includes the pair-versus-normal source signs in Eq. (19); equivalently, those signs may be extracted and the remaining susceptibility denoted by χ . The tables above diagonalize K_G , not necessarily $\mathcal{R}$. Unless the two matrices commute, a kernel eigenvector is not an eigenvector of J_G .

Symmetry still imposes substantial block structure:

- At a number-conserving reference, anomalous pair fields and normal Hartree/exchange fields do not mix to linear order. They can mix after a paired state breaks $U(1)$.
- At an unpolarized spin-symmetric reference, charge-like and spin-like normal perturbations form separate blocks.
- Exact leg reflection separates $k_y = 0$ and $k_y = \pi$ blocks.
- Exact translation would separate longitudinal momenta q_x . The finite open ladder has no exact translation symmetry, so nearby standing waves and boundary-localized profiles can mix weakly or strongly.

Thus “block diagonal” is the appropriate expectation, not “diagonal.” Within one block, on-site, rung, leg-bond, longer-range, Hartree, and exchange coordinates may mix whenever they share all exact quantum numbers. In particular, extended s - and d -wave pair templates are not distinct irreducible representations of an anisotropic two-leg ladder: both are leg-even, spin-singlet, $q_x = 0$ fields. Their rung/leg relative sign is an output to diagnose, not a symmetry prohibition against mixing.

Isolated-ladder intuition. Introduce sign-adjusted conjugate operators $\tilde{O}_a = -\partial H / \partial x_a$, so every source is written as $-x_a \tilde{O}_a$. At zero temperature their static susceptibility has the spectral form

$$\chi_{ab} = 2 \text{Re} \sum_{m>0} \frac{\langle 0 | \tilde{O}_a | m \rangle \langle m | \tilde{O}_b | 0 \rangle}{E_m - E_0}. \quad (21)$$

An off-diagonal entry is nonzero when two different source operators couple the ground state to the same excited states. This is generic on a ladder. A leg-even density potential changes not only the density but also kinetic-energy and exchange-bond expectation values; conversely, a hopping-like β_σ perturbation redistributes charge. Schematically,

$$\begin{pmatrix} \delta n \\ \delta X \end{pmatrix} = - \begin{pmatrix} \chi_{nn} & \chi_{nX} \\ \chi_{Xn} & \chi_{XX} \end{pmatrix} \begin{pmatrix} \delta \nu \\ \delta \beta \end{pmatrix}. \quad (22)$$

The same effect occurs in the pair block: rung hopping t_0 , longitudinal hopping, and interactions allow a leg-bond pair source to induce a rung-pair correlator and vice versa. Reciprocity makes the properly normalized susceptibility symmetric in the exact calculation, but the product $J_G = K_G \mathcal{R}$ need not be symmetric because the geometry kernel weights the output coordinates differently.

A two-coordinate pair example. As a schematic example in the leg/rung basis, suppose the isolated ladder has the sign structure

$$\chi_{\text{pair}} = \begin{pmatrix} 1 & -0.6 \\ -0.6 & 1.2 \end{pmatrix}. \quad (23)$$

The negative off-diagonal response means that a positive leg source induces an opposite-sign rung response, as expected for a d -like tendency. For either cubic geometry,

$$\frac{J_{\text{cubic}}}{g} = \begin{pmatrix} 6 & 0 \\ 0 & 4 \end{pmatrix} \begin{pmatrix} 1 & -0.6 \\ -0.6 & 1.2 \end{pmatrix} = \begin{pmatrix} 6 & -3.6 \\ -2.4 & 4.8 \end{pmatrix}. \quad (24)$$

This matrix is not diagonal. Its eigenvalues are $8.4g$ and $2.4g$; the leading right eigenvector is proportional to $(1, -2/3)^\top$, not to either a pure leg field, a pure rung field, or the equal-amplitude seed $(1, -1)^\top$. It is the field pattern that the complete raw map reproduces, multiplied by $8.4g$.

For the same illustrative ladder response, the square kernel gives

$$\frac{J_{\text{square}}}{g} = \begin{pmatrix} 2 & 0 \\ 0 & 0 \end{pmatrix} \chi_{\text{pair}} = \begin{pmatrix} 2 & -1.2 \\ 0 & 0 \end{pmatrix}. \quad (25)$$

The leg source can still induce a rung correlator through χ_{pair} , but the square kernel removes that output before it becomes the next rung field. The leading self-reproducing field direction is consequently pure leg in this toy projection.

What the current Stage 2 calculation measures. Let the columns of Q be orthonormal candidate fields. The finite-field calculation measures every column of

$$Z = \mathcal{R}Q, \quad \mathcal{R}Q = Q^\top Z, \quad J_{G,Q} = Q^\top K_G Z. \quad (26)$$

Here $Q^\top$ denotes projection using the implemented field metric. It then diagonalizes the complete projected normal and pair blocks, rather than treating the named candidates as independent scalar eigenmodes. The present bank contains nine normal Hartree directions and three independent pair directions spanning the on-site, rung, and leg templates; the named extended s and d templates are exact combinations inside that pair span.

This is a *projected* Jacobian, not yet the literal full array-space Jacobian. In particular, the current normal input basis has no independent β_σ source directions. Exchange-field output is retained as a β -fraction/leakage diagnostic. If that leakage is appreciable, the normal basis must be enlarged with exchange directions before its eigenvalues can be called full normal-sector stability eigenvalues. The same principle applies to response outside the selected longitudinal modes or retained pair templates.

For a linearized mode,

$$\delta x_{m+1} = \lambda \delta x_m. \quad (27)$$

A negative λ flips the mode's sign on every raw update. If $|\lambda| < 1$, this is merely a decaying oscillation into the normal fixed point. If $|\lambda| > 1$, the normal state is unstable along a sign-alternating trajectory; nonlinear saturation can then produce two fields that are exchanged by the raw map. In the one-ladder formulation these two iterations can represent transverse sublattices A and B , provided their actual neighbor maps implement $X_A = R(X_B)$ and $X_B = R(X_A)$. Here

R is the raw one-subsystem response including field construction. The two-step composition $R \circ R$ has linear factor λ^2 at the identical-subsystem fixed point. This is distinct from the *one-sweep* simultaneous two-sublattice Jacobian,

$$J_{\text{cell}} = \begin{pmatrix} 0 & J \\ J & 0 \end{pmatrix}, \quad \lambda_{\text{even}} = \lambda, \quad \lambda_{\text{odd}} = -\lambda. \quad (28)$$

Consequently, for this same-map bipartite extension, complete raw-map linear stability $\rho(J) < 1$ also implies stability to both spatial parities. Observed relaxation from a few seeds is weaker evidence: it does not measure every eigenmode or exclude a finite-amplitude competing basin. This spatial interpretation requires the actual connectivity to support the assignment; numerical period two alone is not sufficient. A converged two-cycle is a stationary state of the spatial cell map, not physical time dependence.

Leg parity is not ladder-sublattice parity. For two spatial ladders with legs 0, 1, the charge deviations $(\rho, -\rho; \rho, -\rho)$ and $(\rho, \rho; -\rho, -\rho)$ are different modes. The first is leg-odd within each ladder; the second preserves each rung's internal charge symmetry while alternating between ladders. The solver retains independent site densities, so the first mode is allowed even when rung-averaged plots hide it. In the rung bonding/antibonding basis, $n_0 - n_1$ mixes the two sectors, whereas $n_0 + n_1$ is their total occupation. Strong rung hopping can therefore give the two charge channels very different responses. This qualitative argument does not substitute for a measured relative-charge gap or a dedicated leg-odd perturbation. Leg-odd spin is a separate response channel.

Trellis repetition requires an additional check. For equal zigzag hoppings, the skew one-ladder map uses a backward hopping kernel on one target leg and a forward kernel on the other. The rectangular two-ladder map uses forward kernels on both legs of A , backward kernels on both legs of B , and a physical $-1/2$ -rung origin for B . A skew two-cycle and a rectangular spatial cell therefore impose different registration conditions on a longitudinal stripe. In the bulk coordinates $x = i - m/2$, the repeated skew cell ties longitudinal modulation to transverse translation; two skew repetitions translate by $(-1, 2)$, whereas the rectangular cell repeats by $(0, 2)$. Open ends also differ. Both cells retain both zigzag bonds and the frustrated triangles. Enlarging the cell does not remove frustration, and the equivalence in Eq. (28) cannot be assumed between these two different trellis maps.

3.2.3 Why the geometries promote d -wave pairing differently

For the two leading bond classes, collect the leg-even nearest-neighbor leg-bond correlator and the even rung-pair correlator into $\mathbf{F}_d = (F_{\parallel,+}, F_{\perp,+})^\top$. A conventional ladder d -wave form factor has opposite signs on these two bond classes, $F_{\parallel,+} F_{\perp,+} < 0$. Although the kernel does not mix different rung pairs, its action on this assembled form-factor vector is

$$K_{\text{frustrated}}^{(d)} = K_{\text{unfrustrated}}^{(d)} = \begin{pmatrix} 6g & 0 \\ 0 & 4g \end{pmatrix}, \quad K_{\text{square}}^{(d)} = \begin{pmatrix} 2g & 0 \\ 0 & 0 \end{pmatrix}. \quad (29)$$

Both cubic geometries therefore provide exactly the same *direct* feedback to the d -wave sector. They preserve the relative minus sign, weight the leg-bond component by $6g$, and weight the rung component by $4g$. On the common leg-bond component this is three times the square feedback.

More importantly, when the ladder's pair structure is rung-singlet dominated, as in the strong-rung $t_0 > 1$ regime of interest, the square kernel discards the dominant component: it returns no rung-pair field at all. In the square connectivity, the two legs of a rung tunnel toward different neighboring ladders, so the rung pair cannot make the same second-order pair hop into one neighboring ladder.

Zero rung *feedback* does not imply a zero rung-pair correlator. The isolated ladder can still convert a leg-bond source into a rung response through off-diagonal entries of its pair susceptibility. The missing square-kernel entry means that this rung response is not fed back as a rung source on the next SCF step.

At a common bare isolated-ladder expansion point, the response matrix χ_{pair} is geometry independent, and

$$J_G^{(d)} = K_G^{(d)} \chi_{\text{pair}}. \quad (30)$$

The displayed 2×2 matrices summarize bond-class weights. The full susceptibility may also mix different longitudinal separations and the on-site pair component, but the equality of the two cubic kernels persists throughout the leg-even pair block. Hence the two cubic geometries have identical bare Jacobians within the leg-even d -wave subspace; any difference between their final d -wave solutions must enter indirectly through competing normal channels, geometry-dependent normal-state dressing, or nonlinear selection. Equation (29) also shows that there is generally no single scalar “ d -wave kernel eigenvalue”: the leg and rung coordinates have different weights, and the full eigenvector and eigenvalue come from $K_G^{(d)} \chi_{\text{pair}}$. The shorthand $\lambda = k\chi$ is exact only for an unmixed eigenmode.

The two cubic geometries separate most clearly in the leg-odd sectors. The same-leg/local $k_y = \pi$ kernel is $-6g$ in the unfrustrated geometry but only $+2g$ in the frustrated geometry: the magnitude is reduced by a factor of three and the sign is reversed. For a leg-odd pair field this is a competing pair channel, not the conventional leg-even d -wave channel. For a leg-odd Hartree or exchange field, the extra minus response sign in Eq. (19) gives the opposite transverse interpretation: the unfrustrated mode is transversely in phase, whereas the weaker frustrated mode wants to alternate between neighboring ladders. This leg-odd sector is the natural leg-staggered SDW/CDW competitor.

There is a second, distinct frustration effect in the leg-even normal sector. Both cubic kernels have $k = +6g$, so the normal response gives $\lambda < 0$ and favors an A/B alternation between neighboring ladders. The bipartite unfrustrated transverse lattice can realize that pattern; the non-bipartite frustrated lattice cannot satisfy it on every bond. Thus “leg-odd” ($k_y = \pi$ within one ladder) and “transversely staggered” (negative SCF λ) are independent labels and should not be conflated. More generally, no negative- λ channel has a globally consistent two-sublattice realization on the frustrated connectivity; a numerical period-two orbit there is not automatically a physical state.

The remaining odd pairing channel is also informative but is not part of a uniform rung d -wave form factor. The diagonal-only cross-leg odd mode has kernel eigenvalues $+4g$, $-4g$, and 0 in the frustrated, unfrustrated, and square geometries, respectively. It vanishes for a rung pair by Eq. (16).

Direct comparison with CDW and SDW. Define the charge- and spin-like Hartree sources and responses by

$$\nu_c = \frac{\nu_\uparrow + \nu_\downarrow}{\sqrt{2}}, \quad \nu_s = \frac{\nu_\uparrow - \nu_\downarrow}{\sqrt{2}}, \quad \chi_c \neq \chi_s \quad \text{in general.} \quad (31)$$

The geometry kernel is spin blind, so CDW and SDW fields with the same leg parity have the same k . Their physical responses differ through the longitudinal wavevector and the isolated-ladder susceptibilities χ_c and χ_s . With hole density $\delta = 1 - n_*$, the principal candidates used in this project are a leg-even CDW near $q_x \simeq 2\pi\delta$ (the $4k_F$ wavevector modulo 2π) and a leg-odd SDW near $q_x \simeq \pi(1 - \delta)$. Table 2 compares these with uniform d -wave pairing.

Tendency / component	Field and ladder sector	Frustrated cubic	Unfrustrated cubic	Square
d -wave, leg bond	$\alpha_{\parallel,+}, q_x = 0, k_y = 0$	$+6g$	$+6g$	$+2g$
d -wave, rung bond	$\alpha_{\perp,+}, q_x = 0, \text{cross-leg even}$	$+4g$	$+4g$	0
CDW candidate	$\nu_c, q_x \simeq 2\pi\delta, k_y = 0$	$-6g$	$-6g$	$-2g$
SDW candidate	$\nu_s, q_x \simeq \pi(1 - \delta), k_y = \pi$	$-2g$	$+6g$	$+2g$

Table 2: Signed geometry factors entering the linearized response. Pair rows use $+k$, while CDW/SDW rows already include the minus source-response sign and therefore use $-k$. Each entry must still be multiplied by its own ladder susceptibility, and the two d -wave components generally mix.

The table makes the relative mechanisms transparent:

- In the frustrated cubic geometry, d -wave feedback remains strong and positive. The leading leg-even CDW factor has the same magnitude $6g$, but its negative sign calls for a transverse two-sublattice pattern that is incompatible with the non-bipartite connectivity. The leg-odd SDW factor is both three times weaker than in the unfrustrated cubic geometry and negative, so its simple transverse alternation is frustrated as well.
- In the unfrustrated cubic geometry, the d -wave leg component, the charge-even CDW, and the spin-odd SDW all carry geometry factors of magnitude $6g$. The CDW factor is negative and its transverse A/B pattern is allowed; the SDW factor is positive and is in phase across ladders. Pairing therefore receives no comparable relief from its main normal competitors, and the winner is controlled primarily by χ_{pair} , χ_c , and χ_s .
- In the square geometry, the common leg scale is only $2g$, so both pairing and normal tendencies are weakened relative to cubic. The decisive asymmetry is that CDW and SDW feedback remains nonzero and can use the bipartite transverse pattern where needed, whereas the dominant rung component of d -wave pairing receives exactly zero direct feedback.

These are Hartree-channel comparisons. A complete CDW or SDW eigenmode may also contain the charge- or spin-like combination of β_σ . Because exchange and Hartree perturbations can mix, the quantitative comparison is between the spectral radii of the full pair, charge, and spin Jacobian blocks, not between isolated entries of Table 2. Geometry fixes the feedback weights and transverse sign constraints; the ladder response decides how strongly each longitudinal order parameter exploits them.

When the isolated ladder is spin gapped, the gap provides an additional, geometry-independent suppression of the low-energy SDW response. It can favor singlet pairing relative to SDW, but it does not distinguish the three transverse geometries; that distinction still comes from the kernels above. The pairing–CDW competition can therefore remain the closer one. Equal-time correlation amplitudes or decay exponents are useful diagnostics but cannot be substituted for the static susceptibilities in the Jacobian comparison.

The resulting kernel-level expectation is that `cubic_frustrated` has the strongest net tendency to stabilize d -wave order: it retains the same $6g/4g$ pair feedback as `cubic_unfrustrated` while weakening or geometrically frustrating key normal competitors. The unfrustrated cubic geometry comes next, and the square geometry is least favorable because it has only $2g$ leg feedback and no rung feedback. This is a mechanism and ranking hypothesis, not by itself a phase-boundary result. A dressed susceptibility can depend indirectly on geometry, and a non-bipartite array may support multi-sublattice or complex orders outside the present real one-/two-sublattice description. Final claims must use the measured full Jacobian together with converged variational-energy comparisons.

The September 18 nonlinear runs illustrate the limitation of this kernel-level expectation: both tested starts reach stripes throughout the unfrustrated-cubic grid, including two coordinates where square remains paired. Stronger bare pair feedback does not by itself imply a larger paired region when the normal competitors and the dressed ladder response change. The dated results are discussed in `docs/manuscript/introduction_and_results.tex` and `docs/reports/campaign_review_20260918/README.md`.

4 MPS ground-state solve and density targeting

4.1 Finite-system DMRG

For fixed x_k and μ , the ground state of H_{eff} is approximated by finite-system DMRG using ITensorMPS. A cold start uses a randomized product state with particle count nearest $2Ln_*$; subsequent MF and chemical potential solves warm-start from the previous MPS. The observer records each sweep energy, maximum discarded weight, and maximum link dimension. It may terminate only after reaching the final configured bond dimension with zero noise and satisfying the sweep-energy tolerance, or when the wall-time deadline is reached.

Production GPU calculations use dense tensors in Float64, with S_z and fermion-parity quantum-number block structure disabled for performance. This changes the tensor representation, not the Hamiltonian. Separate fixed-sector gap calculations use QN-conserving CPU MPS states.

4.2 Safeguarded chemical-potential search

Each MF update contains an inner solve of

$$h(\mu) = n(\mu; x_k) - n_* = 0. \quad (32)$$

The algorithm first evaluates the previous μ . If a positive carried slope $\kappa \simeq dn/d\mu$ is available, it tries a bounded Newton predictor. Otherwise, or if that predictor does not bracket the target, it expands a bracket geometrically in the density-correct direction. Within a bracket it uses a safeguarded secant step, rejecting points within ten percent of either endpoint in favor of bisection. Every trial reuses the nearest MPS and a small warm-start noise schedule. The best-density state is returned on an iteration limit, but it cannot pass final acceptance unless the configured density gate is met.

4.3 Current limitation: equal ladder densities and parallel stripes

The spatial A/B extension still calls `find_mu_for_density` independently for each ladder with the same target $n_\star$. Thus it enforces

$$\bar{n}_m = \frac{1}{2L} \sum_{i,\ell} \langle n_{m,i,\ell} \rangle = n_\star \quad \text{on every ladder } m, \quad (33)$$

rather than only the total cell density. Independent spatial profiles do not release this constraint. Alternating hole-rich and hole-poor ladders, $\bar{n}_A = n_\star + \Delta n$ and $\bar{n}_B = n_\star - \Delta n$, are excluded. Such charge redistribution can produce stripes parallel to the ladders, with modulation across the array. Longitudinal density waves with different A/B phases remain allowed. The excluded mode is the $q_\parallel = 0$, nonzero- $q_\perp$ component of the ladder-average charge; transverse leg redistribution that preserves each average is a different channel.

Literature and relevance. Resonant scattering in the copper-oxide ladder family found hole crystals with periods five and three *along* the ladders [5, 6]. Coherence across neighboring ladders does not imply unequal average ladder densities. These experiments support the retained longitudinal ordering channel, but do not test the excluded one. The related DMRG plus interladder mean-field study of Almeida et al. develops longitudinal CDWs from screened Coulomb coupling, including inequivalent period-four ladder profiles, within specified patterns about a common mean density [7]. It is not an unrestricted charge-transfer stability calculation.

Parallel stripes nevertheless have a numerical motivation: anisotropic two-dimensional t - J studies find a preference for the stronger-coupling direction in low-doping regimes [8, 9]. The iPEPS study also finds an orientation reversal at higher doping. These models have uniform directional anisotropy rather than imposed bulk charge modulation. They establish a plausible competitor, not its preference in our weakly coupled two-leg units or frustrated trellis geometry. Detailed distinctions between isolated ladders, finite-width cylinders, coupled-ladder calculations and two-dimensional orientation studies are in the manuscript’s section “Current limitation: equal ladder densities and parallel stripes.”

Conditional energetic argument. For identical decoupled ladders with smooth energy per site $e(n)$,

$$e(n_\star + \Delta n) + e(n_\star - \Delta n) - 2e(n_\star) = e''(n_\star)(\Delta n)^2 + O((\Delta n)^4). \quad (34)$$

Positive inverse compressibility penalizes redistribution; interladder energy gains and spin reconstruction must overcome it. Neither a spin gap nor strong longitudinal CDW correlations alone imply this instability. No endpoint curvature or charge-transfer stability has been measured here; the expansion is insufficient at nonanalyticities or for a competing finite-amplitude minimum.

Interpretation and prospective control. Current phase comparisons, including accepted square A/B paired states, are restricted to equal average ladder densities. A future test would use one common external chemical potential for the total cell density, allow independent ladder densities and local Hartree fields, and compare imbalance seeds with consistent density gates and canonical energies. Small seeds test only local stability. Transverse size is an additional restriction: at $p = 1/16$, illustrative stripe fillings of one-half and one hole per longitudinal lattice spacing imply spacings of four and eight two-leg ladders on square, respectively, from $p = \rho_\ell / \lambda_\perp$. Antiphase spin order can require a doubled repeat. Thus a negative A/B imbalance test would not exclude longer-period parallel stripes. These are proposed controls, not implemented changes or new simulations.

5 Self-consistency, mixing, and recurrence

One outer iteration is:

1. solve $H_{\text{eff}}[x_k, \mu_k]$ by density-targeted DMRG;
2. measure $y_k = f(x_k)$ and the associated correlators;
3. evaluate field residuals and the variational-energy identities;
4. classify the recent history as a fixed point, physical orbit, candidate, or incomplete numerical trajectory; and
5. if nonterminal, choose x_{k+1} by the raw map or an accelerator.

The field residuals use the complete stored field vector,

$$\epsilon_\infty = \max_j |(y_k - x_k)_j|, \quad \epsilon_2 = \frac{\|y_k - x_k\|_2}{\max(\|y_k\|_2, \|x_k\|_2, \epsilon_{\text{mach}})}. \quad (35)$$

5.1 Unmixed probe and physical orbits

The production-shaped workflow begins with complete raw updates $x_{k+1} = f(x_k)$, before damping or Anderson acceleration. A period- q recurrence requires every phase to close over the configured number of repeat links. With three repeat links, a period-two test needs eight contiguous raw records. Acceptance additionally requires:

- raw-map closure $x_k \simeq y_{k-1}$ for every stored link;
- target density and per-phase energy recurrence;
- the Hamiltonian and effective-eigenvalue consistency checks in Sec. 6; and
- explicit inclusion of q in the configured accepted-period list.

For $q = 2$, successive step vectors must reverse direction: the worst recent cosine must be below its negative threshold, and $\|y_k - y_{k-2}\|/\|y_k - y_{k-1}\|$ must remain below its configured bound. This rejects a slowly drifting trajectory that happens to be close at lag two.

A mixer-dependent recurrence is only a candidate. It is archived, mixer history is cleared, and a fresh raw-map probe is required before it may be accepted. Orbit phases are stored separately and never replaced by their field average. The physical period-two construction follows the logic of Ref. [2]; naming an accepted orbit “CDW” still requires a nonzero bulk density contrast, the expected spatial pattern, and stability with L , bond dimension, and bulk window. The stored contrast is the mean absolute density difference between neighboring orbit phases over a centered bulk fraction of the ladder (one half by default).

5.2 Fixed-point gate and slow modes

A period-one solution must satisfy both the raw residual gate in Eq. (35) and a slow-mode extrapolation. For residual vectors $r_k = y_k - x_k$,

$$\lambda_k = \frac{r_k \cdot r_{k-1}}{\|r_{k-1}\|_2^2}. \quad (36)$$

When consecutive residuals are sufficiently parallel, the gated residual is inflated by

$$s_k = \max\left(1, \frac{1}{1 - \lambda_k}\right). \quad (37)$$

If $\lambda_k \geq 1$, the factor is infinite. This prevents a nearly marginal mode from being accepted merely because its instantaneous residual crosses a local threshold. The recent stability window must also pass the density gate; the latest target-density-corrected energy change and the current energy identities must pass their respective gates.

5.3 Linear and Anderson acceleration

If the initial raw probe finds no accepted solution, the code uses adaptive linear or Anderson mixing. Linear mixing is

$$x_{k+1} = x_k + \eta_k [f(x_k) - x_k]. \quad (38)$$

For Anderson mixing, recent residuals $r_j = f_j - x_j$ are combined with coefficients c_j , constrained by $\sum_j c_j = 1$, by solving the regularized Gram/KKT system. The update is

$$x_{k+1} = (1 - \eta_k) \sum_j c_j x_j + \eta_k \sum_j c_j f_j. \quad (39)$$

The damping is halved after substantial residual growth and increased conservatively after improvement, within configured minimum and maximum bounds. Mixing accelerates a fixed-point search; it does not certify a physical orbit.

5.4 Initial-condition protocol

Independent branches deliberately sample multiple basins. The currently implemented seed families are summarized below.

Seed family	Purpose and construction
Legacy pairing/SDW/CDW	Compatibility protocol: random retained real α , or deterministic staggered Hartree fields. The channels do not have equal seed norms.
Matched mode	Uses a declared rung profile $g_m(i) = \cos\{\pi[m(i-1)/(L-1) + \phi_\pi]\}$, mean-controlled for nonzero m , and normalizes the complete field vector so $\ x\ _2/\sqrt{2L} = A_{\text{seed}}$.
Pairing form factors	On-site s , rung s , leg s , extended s , and d -wave templates, optionally with a declared longitudinal mode.
Stripe	Combines a longitudinally and transversely staggered SDW envelope m with a leg-even charge second harmonic $2m$.
Stripe+pairing	Adds a separately normalized pairing component to test unrestricted coexistence; a pure stripe can remain trapped in the invariant $\alpha = 0$ subspace.
Legacy-like pairing	Translation-invariant random relative-bond α coefficients, with $\beta = \nu = 0$, reproducing the geometry of the old fresh-run source at the matched total norm.

Different seeds are initial conditions, not phase labels. All accepted distinct basins are retained for the variational comparison.

6 Zero-temperature variational functional

6.1 Canonical energy and the particle-number constraint

The branch comparison uses the zero-temperature canonical energy of the coupled-ladder MF approximation,

$$E_{\text{var}}[\psi] = \langle H_{\text{lad}} \rangle_\psi + E_\perp[\psi], \quad \langle N \rangle_\psi = N_\star = 2Ln_\star. \quad (40)$$

The bare Hamiltonian H_{lad} contains no $-\mu N$ term. It supplies the intraladder kinetic and interaction energies, while $E_\perp$ supplies the transverse MF interaction energy. These are total energies per representative ladder; the energy per physical site is $e = E/(2L)$. There is no finite-temperature entropy term.

The chemical potential is a Lagrange multiplier used in the constrained solve:

$$\mathcal{L}[\psi, \mu] = E_{\text{var}}[\psi] - \mu(\langle N \rangle_\psi - N_\star). \quad (41)$$

Variation with respect to the state gives the linear MF Hamiltonian $H_{\text{eff}} = H_{\text{lad}} - \mu N + H_{\text{lin}}$; the scalar $\mu N_\star$ does not affect its eigenvectors. Thus the auxiliary Hamiltonian contains $-\mu N$, even though the energy being compared is canonical. Because the anomalous MF terms allow particle-number fluctuations, the implemented constraint fixes $\langle N \rangle$; it does not project the MPS onto an exact N -particle sector.

6.2 Transverse energy and double-counting correction

The DMRG eigenvalue E_{eig} of H_{eff} is not directly comparable between MF branches: it includes $-\mu N$ and only the linearized transverse terms. Let $L_{\text{lin}} = \langle H_{\text{lin}} \rangle$. Channel by channel, using a collective index a for each channel's retained spatial and spin entries (with on-site exchange excluded),

$$L_{\text{pair}} = -2 \sum_a \alpha_a F_a, \quad E_{\perp, \text{pair}} = \frac{1}{2} L_{\text{pair}}, \quad (42)$$

$$L_{\text{ex}} = \sum_a \beta_a X_a, \quad E_{\perp, \text{ex}} = \frac{1}{2} L_{\text{ex}}, \quad (43)$$

$$L_{\text{dens}} = \sum_a \nu_a n_a, \quad E_{\perp, \text{dens}} = \frac{1}{2} \sum_a \nu_a (n_a - \frac{1}{2}). \quad (44)$$

The applied partner fields are used in these contractions. At a fixed point they equal the outgoing fields; on an orbit they belong to the preceding phase and must remain distinct.

The factor of one half restores the interaction energy from its linear MF terms. For example, for commuting operators on two different ladders and $m_A = \langle O_A \rangle$, $m_B = \langle O_B \rangle$, the decoupling is

$$JO_A O_B \longrightarrow J(m_B O_A + m_A O_B - m_A m_B). \quad (45)$$

The two linear terms have expectation $2Jm_A m_B$, but the bond energy is $Jm_A m_B$. Assigning half of each transverse bond to each ladder yields the channel energies above. The density channel additionally retains the reference shift $\delta n = n - \frac{1}{2}$: its physical MF energy is $\frac{1}{2} \sum_a \nu_a \delta n_a$, although its linear Hamiltonian term has expectation $\sum_a \nu_a n_a$.

With $E_{\perp}$ the sum of the three quadratic MF contributions, the double-counting constant is

$$C_{\text{dc}} = E_{\perp} - L_{\text{lin}}. \quad (46)$$

The reconstructed canonical functional is

$$E_{\text{var}}^{\text{rec}} = \langle H_{\text{eff}} \rangle + \mu \langle N \rangle + C_{\text{dc}}, \quad (47)$$

and the independently evaluated direct form is

$$E_{\text{var}}^{\text{dir}} = \langle H_{\text{lad}} \rangle + E_{\perp}. \quad (48)$$

The code stores Eq. (48) as the canonical variational energy and requires agreement with the reconstructed form. It also checks

$$\langle H_{\text{lad}} \rangle = \langle H_{\text{eff}} \rangle + \mu \langle N \rangle - L_{\text{lin}} \quad (49)$$

and separately compares E_{eig} with $\langle H_{\text{eff}} \rangle$.

The cancellation can be written explicitly as

$$\begin{aligned} E_{\text{var}} &= \underbrace{\langle H_{\text{lad}} \rangle - \mu \langle N \rangle + L_{\text{lin}}}_{\langle H_{\text{eff}} \rangle} + \mu \langle N \rangle + (E_{\perp} - L_{\text{lin}}) \\ &= \langle H_{\text{lad}} \rangle + E_{\perp}. \end{aligned} \quad (50)$$

At exactly the same target particle number, two branches A and B therefore have the canonical energy difference

$$\Delta E_{\text{var}} = (\langle H_{\text{lad}} \rangle_A - \langle H_{\text{lad}} \rangle_B) + (E_{\perp,A} - E_{\perp,B}), \quad (51)$$

with no explicit chemical-potential term. Equal particle number alone does not cancel the auxiliary terms in the raw effective energies: different branches can require different μ_A and μ_B , leaving $-N_*(\mu_A - \mu_B)$ in their raw difference. Each branch's $-\mu \langle N \rangle$ contribution must be removed separately, along with the linear-field double counting.

6.3 Residual density correction and branch ranking

Density targeting is numerical, so branch b may end at $N_b = \langle N \rangle_b$ slightly different from N_* . Along a smooth self-consistent branch, its constraint multiplier is the local slope $\mu_b = \partial E_b / \partial N$. Taylor expansion gives

$$E_b(N_*) = E_b(N_b) + \mu_b(N_* - N_b) + O((N_* - N_b)^2). \quad (52)$$

The stored comparison energy uses this leading interpolation,

$$E_{N_*} = E_{\text{var}}^{\text{dir}} + \mu(N_* - \langle N \rangle). \quad (53)$$

Equation (53) is used for energy-stability gates and branch ranking. This correction is applied after reconstructing the canonical energy; it is distinct from adding back $\mu \langle N \rangle$ to remove the auxiliary Hamiltonian's chemical-potential term. It vanishes at exact target density and neither changes the MPS nor makes its particle number exact. The interpolation assumes a smooth branch near N_* ; it is not a substitute for tighter density convergence or a certified error bar.

For a validated physical period- p orbit, the comparison energy is

$$\overline{E}_{N_*} = \frac{1}{p} \sum_{j=1}^p \left[\langle H_{\text{lad}} \rangle_j + E_{\perp,j} + \mu_j(N_* - \langle N \rangle_j) \right]. \quad (54)$$

Each phase uses its applied partner fields in $E_{\perp,j}$; the arithmetic average is taken over energies, not over fields or MPS states. A fixed point is the $p = 1$ case. The lower comparison energy selects the favored branch among the eligible solutions tested; the difference includes every ladder and transverse channel, not only the pairing contribution.

Only accepted fixed points or validated unmixed orbits may be ranked. The model, geometry, numerical settings, implementation, and E_p -registry fingerprints must match. Cross-geometry energies are not ranked because the geometries define different effective Hamiltonians, and a possible field-independent perturbative offset has not been restored.

7 Observables and validation diagnostics

7.1 Structure factors and K_ρ

For $O_i = n_i$ or S_i^z , the connected structure factor is

$$S_O(q_x, q_y) = \frac{1}{2L} \sum_{a,b} e^{i[q_x(i_a - i_b) + q_y(\ell_a - \ell_b)]} (\langle O_a O_b \rangle - \langle O_a \rangle \langle O_b \rangle), \quad (55)$$

with $q_x = 2\pi m/L$ and $q_y \in \{0, \pi\}$. The dominant charge and spin wavevectors are stored. The reference $q_x = \pi n$ is a tracking diagnostic, not a universal two-leg-ladder phase criterion.

The first up to three nonzero $q_y = 0$ charge modes are fit through the origin to obtain $s = dS_c/dq$. Both normalizations are reported,

$$K_{\rho, \text{site}} = \pi s, \quad K_{\rho, \text{rung}} = 2\pi s. \quad (56)$$

7.2 Entanglement and pair correlations

For every MPS bond, Schmidt probabilities p_a give

$$S = - \sum_a p_a \log p_a, \quad S_2 = - \log \sum_a p_a^2. \quad (57)$$

The central-charge diagnostic uses inter-rung cuts away from the open edges and fits

$$S(\ell) = \frac{c}{6} \log \left[\frac{2L}{\pi} \sin \left(\frac{\pi \ell}{L} \right) \right] + s_0. \quad (58)$$

The fit window, number of points, and R^2 are stored. Finite-size, finite-entanglement, and parity effects can make this estimate unreliable.

Optional full pair diagnostics use the unnormalized singlet operator

$$\Delta_{ab} = c_{a\uparrow} c_{b\downarrow} - c_{a\downarrow} c_{b\uparrow} \quad (59)$$

and retain sign-resolved matrices for rung, leg-0, and leg-1 bonds. Spatial decay and sign structure must be reported; a large local value or finite-MPS pair field alone does not establish superconducting long-range order.

7.3 Fixed-sector gaps

Separate QN-conserving DMRG calculations evaluate

$$\Delta_s = E(N, 2) - E(N, 0), \quad (60)$$

$$\Delta_c = \frac{1}{2} [E(N + 2, 0) + E(N - 2, 0) - 2E(N, 0)], \quad (61)$$

$$E_p^{(h)} = E(N - 2, 0) + E(N, 0) - 2E(N - 1, 1), \quad (62)$$

$$E_p^{(p)} = E(N + 2, 0) + E(N, 0) - 2E(N + 1, 1), \quad (63)$$

where the second argument is $2S_z$. The present implementation assumes an even reference N and an $S_z = 0$ reference sector. Per-sector sweep energies, discarded weights, and maximum link dimensions are stored as convergence evidence.

8 Numerical acceptance and representative controls

The following values describe the checked-in Phase-1 base configuration and are not universal constants. Every run stores its actual values.

Control	Representative value / rule
DMRG	12 sweeps, $\chi = 200$, cutoff 10^{-10} , sweep-energy tolerance 10^{-8} , Krylov dimension 8.
Density search	$ n - n_\star \leq 2 \times 10^{-4}$, at most 16 evaluations, initial bracket step 0.05, growth factor 2.
Fixed-point fields	$\epsilon_\infty \leq 10^{-6}$ or $\epsilon_2 \leq 5 \times 10^{-3}$, including the slow-mode extrapolation, for two stable iterations.
Energy and identities	corrected energy change per site $\leq 10^{-7}$, bare-Hamiltonian identity error per site $\leq 10^{-9}$, and effective eigenvalue error per site $\leq 10^{-6}$.
Raw orbit probe	20 raw updates; inspect periods through 2 in the probe; periods 1 and 2 enabled; three recurrence links per phase; recurrence distance $\leq 2 \times 10^{-6}$ absolute or $\leq 10^{-2}$ relative.
Period-two shape	worst step cosine ≤ -0.5 and worst two-step/one-step ratio ≤ 0.5 .
Anderson mixing	memory 5, regularization 10^{-10} , initial damping 0.5, adaptive range 0.05–0.8.

An accepted solution is therefore a finite- L , finite- χ numerical object. Phase claims require, at minimum, bond-dimension and length scaling, boundary/bulk-window checks, consistent order diagnostics, sector gaps, and seed/basin robustness. The implementation currently excludes complex fields and flux order.

8.1 September 19 completed raw-update campaign and energy-shape diagnostics

The current two-family protocol mixes 95% of one reference correlation profile with 5% of the competing reference, then rebuilds the fields with the target couplings. The references are the stripe at $(t_0, V) = (1, 0)$ and paired state at $(1.4, -0.4)$. The recent cubic grid and finer square cuts use $L = 64$, $\chi = 200$, raw MF updates, no Anderson acceleration, and a 60-evaluation cap. Earlier square anchors, remainder runs and continuations retain their original controls. These are representative campaign settings, not a replacement for the generic algorithms above.

The September 19 local review contains 38 complete 60-step run histories: 18 cubic, 12 finer-square, four positive- V square and four trellis. These contain 2280 cell updates and 2400 individual ladder solves because two trellis runs contain spatial ladders A/B . None is formally accepted. Cubic reaches stripes at all nine coordinates; the finer square cuts retain two seed-dependent texture pairs. All four $(1.2, +0.2)$ square starts lose pairing. The period-eight start retains six unevenly spaced spin-envelope nodes and an endpoint excess $9.3242 \times 10^{-4} t/\text{site}$ over the stripe start; this is a defect/coarsening diagnostic, not an accepted energy ranking. Both trellis one-ladder starts at $(1, 0)$ reach the same paired texture, but both rectangular two-ladder starts retain strong, essentially unpaired stripes. Detailed profiles, failed gates and source checks belong to [docs/reports/campaign_review_20260918/README.md](#).

The two-ladder trellis nonacceptance is an outer-iteration issue: final global relative residuals are 0.1845–0.2085, despite passing final inner-DMRG windows. Successive full-cell measured-field increments have cosine near -0.99995 and norm ratio about 0.975; two-sweep maximum differences are only 3.2–3.5% of one-sweep differences, but remain resolved. This supports slowly damped

alternating relaxation, not an accepted temporal orbit. Spatial A/B labels must not be confused with sweep parity. A future damping test would change the numerical iteration, not identify a physical dynamical frequency. No damping or acceptance change accompanies this analysis. Trellis profiles use stored physical correlations, since its Hartree fields include normal-bond contributions.

A September 20 check evaluates two copies of each saved paired ladder in the same rectangular cell as the striped trials, retaining the measured bare-ladder expectation and recomputing every interaction field. This raises the paired energy by only $2.46378 \times 10^{-6} t/\text{site}$. The striped trials remain 0.002238 – $0.002308 t/\text{site}$ lower in that common functional. Residual-density corrections are below $8.33 \times 10^{-6} t/\text{site}$, and the uncorrected canonical gaps have the same sign and scale. Nonstationarity does not invalidate a trial expectation value; nesting is needed for an automatic inequality between optimized variational families, not for this explicit comparison. This establishes a lower-energy striped trial without certifying a stationary or global minimum. Source and kernel hashes are in `docs/reports/trellis_progress_20260918/same_cell_energy_audit_20260920.toml`.

The two-ladder cell also restricts transverse stripe registration. For a charge harmonic $\delta n_m(x) = \rho \cos(q_x x + m\theta)$, a rectangular n -ladder repetition requires $n\theta = 2\pi p$. A rigid shift of d rungs per ladder therefore requires nd to be an integer multiple of the charge period; the full spin texture must close as well. More generally every ordering wavevector must obey $\vec{Q} \cdot \vec{T} = 2\pi p$ with integer p for the actual cell translation, including trellis shear. The observed nonstationary A/B charge phase offset near 126 – 131 degrees alternates in sign on repeating A/B ; it does not establish a continuing diagonal stripe. Larger cells or carefully defined translated repetitions could test that possibility, but no such runs accompany this analysis.

For a transition cut, plot the full stored target-corrected canonical energy per physical site against the varied Hamiltonian parameter, including coarse endpoints. Label curves by initial seed unless a phase is actually continued through parameter space. The diagnostic endpoint energy does not become an accepted branch energy simply because its last few values are flat. Any signed gap between unaccepted endpoints remains a diagnostic.

Small energy-shape changes can be exposed by subtracting one common affine reference from both curves. In the present plots that reference joins the mean of the two seed energies at each outer endpoint. This preserves seed gaps and slope changes; separate fits to each seed would not do so. Adjacent secants $\Delta(E_{\text{target}}/N_s)/\Delta p$ are interval averages, not derivatives measured at a phase boundary. The sum of the endpoints' last-ten energy ranges divided by Δp diagnoses recent drift only; it is neither a statistical error bar nor a bound on remaining relaxation.

The square $t_0 = 1.4$ cut has a small final-interval downturn in slope between $V = -0.05$ and 0 (about 0.38% for the pairing-seed curve). The $V = -0.4$ cut has nearly regular curvature in t_0 . These observations do not resolve a discontinuous first derivative of a minimizing energy. The two textures at one coordinate per cut motivate a first-order test, but are not parameter-sweep hysteresis or two accepted branches crossing. Also, the interpolated E_p changes $g = t_\perp^2/|E_p|$ along each cut: the energy slope includes the changing transverse functional as well as the explicit bare-ladder parameter. All plotted energies retain the implemented convention of Sec. 6.

The companion spin/pairing cuts use the same twenty source endpoints as the energy curves. Spin is the RMS of physical leg-odd S^z over rungs 6 – 59 ; leg pairing is symmetrized and averaged over legs before its RMS on bonds with left rungs 6 – 58 . A central 17 – 48 spin window distinguishes end-weighted remnants. These anomalous amplitudes are not pair-pair correlations. Full raw and connected terminal correlations are now enabled for both accepted and maximum-iteration states; a retrospective measurement pass over 56 latest branches / 58 spatial MPSs is prepared. Such

measurements retain status, residuals and source hashes and do not imply convergence. No new four-fermion data from that pass are used in the present review.

9 Artifacts, provenance, and branch selection

Each HDF5 state records the complete configuration, applied and measured field history, correlators and energy decomposition, DMRG sweep evidence, model and numerical fingerprints, implementation and registry hashes, software/device metadata, random seed, branch labels, and hash-pinned parent or resume lineage. Rolling latest/best checkpoints are replaceable; final states and orbit artifacts are immutable.

Full production states, including MPS objects, live in scratch storage. Compact stateless mirrors omit MPS tensors but retain the analysis and selection data plus hashes and the full-artifact location. A compact state may be plotted, compared, or used for field-only inheritance, but cannot be used as an MPS restart.

The comparison tool requires matching fingerprints and accepts only gated period-one states or explicitly enabled orbits validated with the raw map. For scans, keep independent, forward, reverse, and metastable continuations as distinct lineages rather than overwriting a higher-energy basin.

10 Code map and maintenance

File	Responsibility
<code>src/MeanField.jl</code>	site ordering, seeds, bare/MF MPO, correlators, and geometry-dependent raw field map
<code>src/Geometry.jl</code>	site/rung indexing, supported transverse geometries, and spin-resolved density kernels
<code>src/Solver.jl</code>	DMRG schedules, density targeting, SCF driver, and checkpoint/diagnostic orchestration
<code>src/Convergence.jl</code>	fixed-point, slow-mode, recurrence, orbit, and failure-status gates
<code>src/Mixing.jl</code>	field vectorization, residual metric, linear mixing, and Anderson acceleration
<code>src/Variational.jl</code>	channel constants, direct/reconstructed canonical energy, and consistency identities
<code>src/Diagnostics.jl</code>	structure factors, K_ρ , entanglement, central charge, pair matrices, and fixed-sector gaps
<code>src/EpRegistry.jl</code>	exact/interpolated pair-binding selection and weak-coupling checks
<code>src/Storage.jl</code> , <code>src/Provenance.jl</code> , <code>src/Selection.jl</code>	immutable artifacts, fingerprints, lineage, compact mirrors, and accepted branch ranking

When the scientific map changes, update this file together with the relevant source and tests. In

particular, any change to the $\alpha/\beta/\nu$ map must be reconciled with $E_{\perp}$, the double-counting correction, and the Hamiltonian-identity tests. Campaign outcomes and command histories remain in `docs/RUN_LOG.md`; these notes should contain only durable methods and clearly labeled representative settings.

Revision log

Date	Revision	Summary
2026-09-23	1.0	Documented the separate ladder-density constraint, excluded parallel stripes, experimental and numerical context, conditional compressibility cost and prospective total-density/transverse-cell controls.
2026-09-18	0.9	Added the current raw two-seed campaign, qualified the kernel-level geometry expectation with the new data, and specified full energy-cut, common-background and secant-slope diagnostics.
2026-09-09	0.8	Distinguished correlators from Hamiltonian fields and expanded the canonical constraint, double counting, residual density correction, and orbit-averaged branch comparison.
2026-09-08	0.7	Identified the published 2023 repulsive-ladder benchmark with the project’s frustrated-cubic mean-field map.
2026-09-02	0.6	Expanded the full-response discussion with symmetry blocks, isolated-ladder mixing, a two-component example, and the exact scope of the current projected Stage 2 Jacobian.
2026-09-02	0.5	Added a signed-response comparison of uniform d -wave, charge-even CDW, and spin-odd SDW tendencies across all three geometries.
2026-09-02	0.4	Added the geometry-by-geometry mechanism for d -wave feedback, separated leg parity from transverse staggering, and recorded the kernel-level ordering tendency and its caveats.
2026-09-02	0.3	Added the exchange-field spectrum, the diagonal-only cross-leg odd channels, spin/charge degeneracies, and the distinction between kernel and full-Jacobian eigenvalues.
2026-09-02	0.2	Added the bond-space interpretation, consolidated pairing and Hartree kernel spectra, and documented channel-dependent response signs.
2026-09-01	0.1	Initial synthesis of the modular model, SCF/DMRG method, variational functional, convergence gates, diagnostics, and provenance rules.

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